



Shore

**Year 11  
Mathematics  
Yearly Examination  
September 2014**

**General Instructions**

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board approved calculators may be used
- Answer Questions 1 – 10 on the Multiple Choice answer sheet provided
- Start each of Questions 11 – 14 in a new writing booklet
- In Questions 11 – 14, show relevant mathematical reasoning and/or calculations
- Write your examination number on the front cover of each booklet to be handed in
- If you do not attempt a question, submit a blank booklet marked with your examination number and “N/A”

Examination Number:

Set:

**Total marks – 70**

**Section I** Pages 3 - 7

**10 marks**

- Attempt Questions 1 – 10
- Each question is worth 1 mark
- Allow about 10 minutes for this section

**Section II** Pages 8 – 12

**60 marks**

- Attempt Questions 11 - 14
- Each question is worth 15 marks
- Allow about 1 hour and 50 minutes for this section

**Note: Any time you have remaining should be spent revising your answers.**

**Section I**

**10 marks**

**Attempt Questions 1 - 10**

**Allow about 10 minutes for this section**

**Use the multiple choice answer sheet.**

1 If  $x = 3y^3 - 2$ , what is the value of  $y$  when  $x = 19$  ?

- (A)  $\sqrt[3]{\frac{17}{3}}$
- (B)  $\sqrt[3]{7}$
- (C)  $\sqrt[3]{21}$
- (D)  $\frac{\sqrt[3]{21}}{3}$

2 What is  $\frac{\sqrt{3}}{5+2\sqrt{3}}$  as a fraction with a rational denominator?

- (A)  $\frac{6+5\sqrt{3}}{37}$
- (B)  $\frac{5\sqrt{3}-6}{37}$
- (C)  $\frac{6+5\sqrt{3}}{13}$
- (D)  $\frac{5\sqrt{3}-6}{13}$

**DO NOT REMOVE THIS PAPER FROM THE EXAMINATION ROOM**

3 What is the domain and range of the function  $y = \sqrt{4 - x^2}$ ?

- (A) Domain  $-2 \leq x \leq 2$ , Range  $0 \leq y \leq 2$
- (B) Domain  $-2 \leq x \leq 2$ , Range  $-2 \leq y \leq 2$
- (C) Domain  $0 \leq x \leq 2$ , Range  $-4 \leq y \leq 4$
- (D) Domain  $0 \leq x \leq 2$ , Range  $0 \leq y \leq 4$

4 What is the gradient of the normal to the curve  $y = \frac{3x^2 + 2}{x^2}$  at  $x = 2$ ?

- (A)  $-2$
- (B)  $-\frac{1}{2}$
- (C)  $\frac{1}{2}$
- (D)  $2$

5 What is the solution of  $2^{x+1} = \frac{1}{64}$ ?

- (A)  $x = -7$
- (B)  $x = -5$
- (C)  $x = 5$
- (D)  $x = 7$

6 What is the derivative of  $(4x^2 - 5)^3$ ?

- (A)  $3(4x^2 - 5)$
- (B)  $3(4x^2 - 5)^2$
- (C)  $24x(4x^2 - 5)^2$
- (D)  $12x^2(4x^2 - 5)^2$

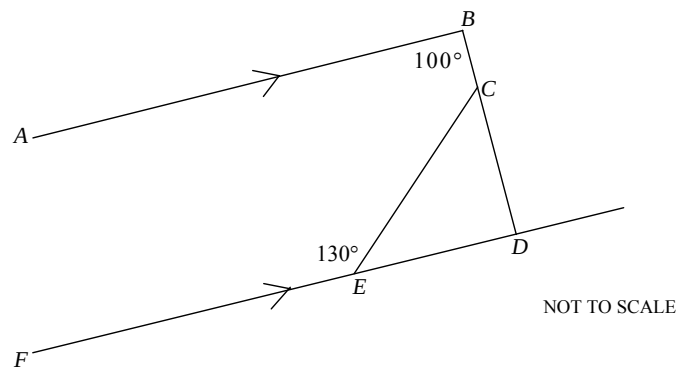
7 What is the simplified expression for  $\frac{x^3 - 1}{x^2 - 1} \times \frac{x^2 - 4x - 5}{4x^2 + 4x + 4}$ ?

- (A)  $\frac{x - 5}{4}$
- (B)  $\frac{x - 1}{4}$
- (C)  $\frac{x + 1}{4}$
- (D)  $\frac{x^2 + x + 1}{4}$

8 What is the solution of the inequality  $|3 - 2x| < 5$ ?

- (A)  $x < -4$  or  $x > 1$
- (B)  $-4 < x < 1$
- (C)  $-1 < x < 4$
- (D)  $x < -1$  or  $x > 4$

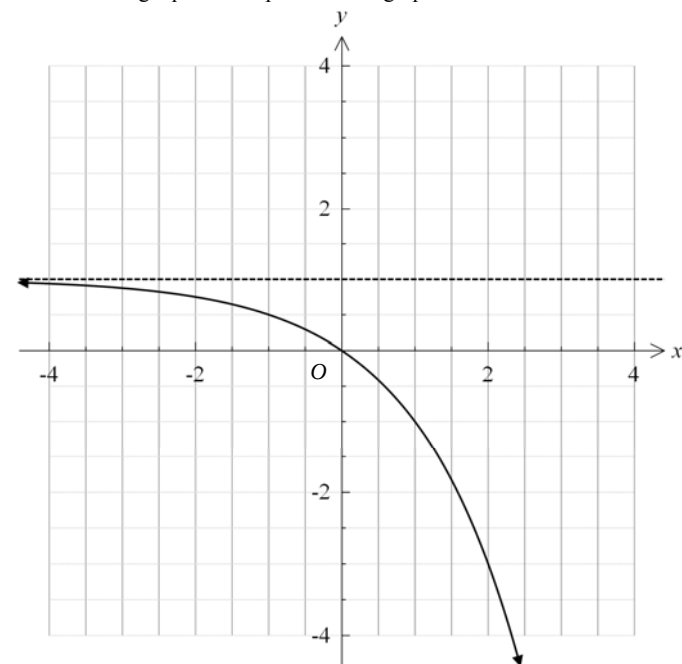
- 9 In the diagram below,  $AB$  is parallel to  $FD$ ,  $\angle ABC = 100^\circ$  and  $\angle CEF = 130^\circ$ .



What is the value of  $\angle BCE$ ?

- (A)  $100^\circ$
- (B)  $110^\circ$
- (C)  $120^\circ$
- (D)  $130^\circ$

- 10 Which of the following equations represents this graph?



- (A)  $y = 2^x - 1$
- (B)  $y = 2^{-x} - 1$
- (C)  $y = 1 - 2^x$
- (D)  $y = 1 - 2^{-x}$

End of Section I

## Section II

**Total Marks 60**

**Attempt Questions 11 - 14**

**Allow about 1 hour and 50 minutes for this section.**

Answer all questions, starting each question in a **new answer booklet** with your exam number clearly visible. Extra writing booklets are available.

In questions 11 – 14, your responses should include relevant mathematical reasoning and/or calculations.

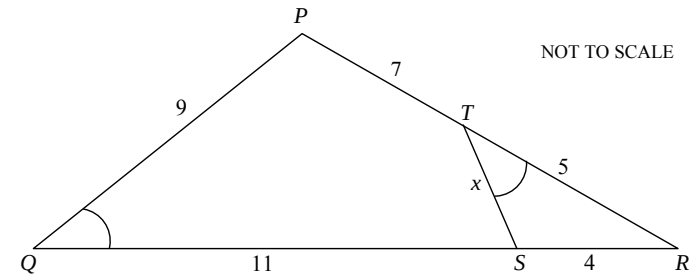
### Question 11 (15 marks)

- (a) A retailer marked up the wholesale price of a jacket by 37% before selling it for \$347.98. Calculate the wholesale price of the jacket to the nearest cent. 2
- (b) Solve  $\frac{x-1}{3} - 1 = \frac{x+2}{2}$ . 2
- (c) What angle does the tangent to the curve  $y = x^3 + x^2$  at  $x = -1$  make with the positive direction of the  $x$ -axis? 2
- (d) Find the size of each interior angle of a regular octagon. 2
- (e) Evaluate  $\lim_{x \rightarrow \infty} \frac{3x^2 - 2x}{3x - 5x^2}$ . 2
- (f) Factorise fully  $x^4 - 16$ . 2

Question 11 continues on the following page

### Question 11 (continued)

(g)



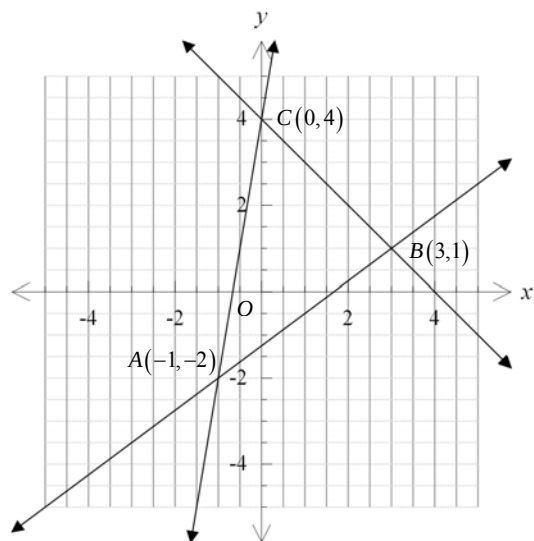
In the diagram,  $\angle PQR = \angle STR$ .

- (i) Prove that  $\triangle PQR$  is similar to  $\triangle STR$ . 2
- (ii) Hence find the value of  $x$ . 1

**End of Question 11**

**Question 12** (15 marks) Use a SEPARATE writing booklet

- (a) Differentiate  $3x\sqrt{x}$ . 2
- (b) If  $y = x^3 - 4x^2 - 5$ , for what values of  $x$  does  $\frac{dy}{dx} = -4$ ? 3
- (c) A triangle has sides 7 cm, 9 cm and 10 cm.
- (i) Find the largest angle of the triangle. Answer to the nearest minute. 2
- (ii) Hence, or otherwise, find the area of the triangle correct to two significant figures. 1
- (d) The diagram below shows the points  $A(-1, -2)$ ,  $B(3, 1)$  and  $C(0, 4)$ .



- (i) Find the centre and radius of the circle which has  $AB$  as diameter. 2
- (ii) Show that the equation of the line through  $A$  and  $B$  is  $3x - 4y - 5 = 0$ . 2
- (iii) Find the perpendicular distance from  $C$  to the line  $AB$ . 2
- (iv) Hence find the area of the triangle  $ABC$ . 1

**Question 13** (15 marks) Use a SEPARATE writing booklet

- (a) If  $\alpha$  and  $\beta$  are the roots of  $2x^2 - 3x - 4 = 0$ , find
- (i)  $\alpha + \beta$  1
- (ii)  $\alpha\beta$  1
- (iii)  $(\alpha - \beta)^2$  2
- (b) Solve  $2\sin 2\theta + \sqrt{3} = 0$  for  $0^\circ \leq \theta \leq 360^\circ$ . 3
- (c) Solve  $|x - 3| = 3x + 1$ . 3
- (d) Find the values of  $k$  for which  $x^2 + kx + 4k = 0$  has real roots. 3
- (e) Prove that  $\frac{1}{\sin^2 \theta} + \frac{1}{\cos^2 \theta} = \sec^2 \theta \operatorname{cosec}^2 \theta$ . 2

**Question 14** (15 marks) Use a SEPARATE writing booklet

(a) Find the equation of the tangent to the curve  $y = (3x + 1)(x - 2)$  which is parallel to the line  $7x - y - 5 = 0$ . **3**

(b) Differentiate  $f(x) = x^2 - 3x$  from first principles. **3**

(c) If  $y = x(2x - 1)^3$ , find  $\frac{dy}{dx}$  in fully factored form. **3**

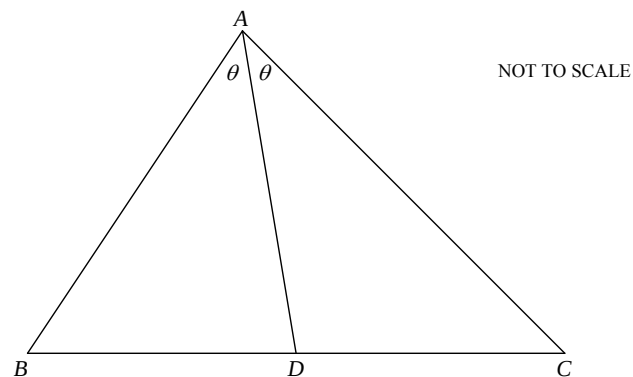
(d) Given the function  $y = \frac{x^2}{x + 1}$ .

(i) Find  $\frac{dy}{dx}$ . **2**

(ii) Find the values of  $x$  for which the tangents to this curve are horizontal. **1**

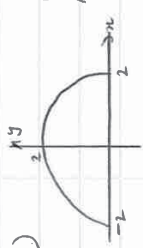
(e) In triangle  $ABC$ ,  $AD$  bisects  $\angle BAC$ . **3**

Use the sine rule to prove that  $\frac{AB}{AC} = \frac{BD}{DC}$ .



**End of Examination**

YEAR 11 MATHEMATICS WORKSHEET SOLUTIONS

- (1)  $x = 3y^2 - 2$   
 $3y^2 = x + 2$   
 $y^2 = \frac{x+2}{3}$   
 $y = \sqrt{\frac{x+2}{3}}$   
 $x=14 \rightarrow y = \sqrt{\frac{14+2}{3}} = \sqrt{6}$
- (2)  $\frac{\sqrt{3}}{5+2\sqrt{3}} = \frac{\sqrt{3}}{5+2\sqrt{3}} \times \frac{5-2\sqrt{3}}{5-2\sqrt{3}}$   
 $= \frac{5\sqrt{3} - 2 \times 3}{25 - 4 \times 3}$   
 $= \frac{5\sqrt{3} - 6}{13}$
- (3)   
 $D: -2 \leq x \leq 2$   
 $R: 0 \leq y \leq 2$
- (4)  $y = 3 + \frac{2}{x^2} = 3 + 2x^{-2}$   
 $\frac{dy}{dx} = -4x^{-3} = -\frac{4}{x^3}$   
 When  $x=2$ ,  $m_t = \frac{dy}{dx} = -\frac{4}{8} = -\frac{1}{2}$   
 $\therefore m_n = 2$
- (5)  $2^{z+1} = 2^{-6}$   
 $\therefore z+1 = -6$   
 $z = -7$
- (6)  $y = (4x^2 - 5)^3$   
 $\therefore y' = 3(4x^2 - 5)^2 \cdot 8x = 24x(4x^2 - 5)^2$
- (7)  $\frac{(x-1)(x^2+x+1)}{(x-1)(x^2+x+1)} \times \frac{(x-5)(x^2+x+1)}{4(x^2+x+1)}$   
 $= \frac{x-5}{4}$
- (8)  $|3-2x| < 5$   
 $-5 < 3-2x < 5$   
 $-8 < -2x < 2$   
 $4 > x > -1 \therefore -1 < x < 4$
- (9)  $\hat{A}BC = 180^\circ - 100^\circ$  (since  $\angle$ 's supplementary)  
 $= 80^\circ$   
 $\hat{A}CB + 80^\circ = 180^\circ$  (Exterior Angle Theorem)  
 $\hat{A}CB = 100^\circ$   
 $\hat{A}CB = 180^\circ - 80^\circ$  (straight angle)  
 $= 100^\circ$
- (10)  $y = 1 - 2^x$
- (11) Let  $\phi$  be whole number  
 $\therefore 137\% \text{ of } \phi = 347.98$   
 $1.37\phi = 347.98$   
 $\phi = \frac{347.98}{1.37}$   
 $= 254$   
 $\therefore \text{whole number} = 254$
- (12)  $\frac{x-1}{3} - 1 = \frac{x+2}{4}$   
 $2(x-1) - 6 = 3(x+2)$   
 $2x-2-6 = 3x+6$   
 $-4 = x$   
 $x = -4$
- (13)  $y = x^2 + x^2$   
 $y' = 2x + 2x$   
 When  $x = -1$ ,  $m_t = 3 - 2 = 1$   
 $\therefore t.m = 1$   
 $\therefore \phi = 45^\circ$
- (14)  $\phi = (n-2) \times 180^\circ$   $n=8$   
 $= 6 \times 180^\circ$   
 $= 1080^\circ$   
 $\therefore \text{each int } \angle = \frac{1080^\circ}{8} = 135^\circ$
- (15)  $2^{z+1} = 2^{-6}$   
 $\therefore z+1 = -6$   
 $z = -7$

- (11) e)  $\text{Let } 3x^2 - 2x = 6x - \frac{3}{x} - 5$   
 $3x^2 - 2x = 6x - \frac{3}{x} - 5$   
 $= -\frac{3}{x}$
- f)  $x^4 - 16 = (x^2 - 4)(x^2 + 4)$   
 $= (x-2)(x+2)(x^2 + 4)$
- g) i)  $\Delta PQR$ ,  $PRQ$ ,  $PRR$   
 $\hat{P}RQ = \hat{S}RT$  (common  $\angle$ )  
 $\frac{QR}{RT} = \frac{RT}{R} = 3$   
 $\frac{PR}{RT} = \frac{RT}{R} = 3$
- $\therefore \Delta PQR \parallel \Delta STR$  (one  $\angle$  equal, sides about  $\angle$  proportional)
- ii)  $\Delta PQR$ ,  $PRR$ ,  $STR$   
 $\hat{P}RQ = \hat{S}RT$  (given)  
 $\hat{Q}RP = \hat{S}RT$  (same  $\angle$ )  
 $\therefore \Delta PQR \parallel \Delta STR$  (Corresponding angles)
- iii)  $\frac{x}{4} = \frac{4}{12}$  (Corresponding sides)  
 $\therefore x = \frac{36}{12} = 3$

$$(12) a) y = 3x\sqrt{x} = 3x \cdot x^{\frac{1}{2}} = 3x^{\frac{3}{2}} \quad (1)$$

$$\frac{dy}{dx} = \frac{3}{2} x^{\frac{1}{2}} = \frac{3}{2} \sqrt{x} \text{ or } \frac{3\sqrt{x}}{2} \quad (1)$$

$$b) y = x^3 - 4x^2 - 5$$

$$\frac{dy}{dx} = 3x^2 - 8x \quad (1)$$

$$\frac{dy}{dx} = -4 \rightarrow 3x^2 - 8x = -4$$

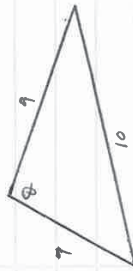
$$3x^2 - 8x + 4 = 0$$

$$(3x-2)(x-2) = 0 \quad (1)$$

$$\therefore 3x-2=0 \text{ or } x-2=0$$

$$x = \frac{2}{3}, 2 \quad (1)$$

c) i) largest angle is opposite largest side



$$\cos \theta = \frac{7^2 + 8^2 - 10^2}{2 \cdot 7 \cdot 8}$$

$$= 0.236095238 \dots \quad (1)$$

$$\therefore \theta = 76^\circ 14' \quad (1)$$

$$ii) A = \frac{1}{2} ab \sin C$$

$$= \frac{1}{2} \cdot 7 \cdot 8 \cdot \sin 77^\circ 31'$$

$$= 30.785 \dots$$

$$= 31 \text{ cm}^2 \quad (2, f) \quad (1)$$

$$(12) i) AB = \sqrt{(3-1)^2 + (1-2)^2}$$

$$= \sqrt{4+1}$$

$$= \sqrt{5} = \sqrt{5} u \quad (1)$$

$$\text{map of AB: } x = \frac{x_2 + x_1}{2} = \frac{3+1}{2} = 2$$

$$y = \frac{y_2 + y_1}{2} = \frac{-2+1}{2} = -\frac{1}{2} \quad (1)$$

$$= 1 = -\frac{1}{2} \quad (1)$$

$$\therefore \text{Centre is } (1, -\frac{1}{2}), \text{ radius} = \frac{5}{2} u$$

$$ii) \frac{y-y_1}{x-x_1} = \frac{y_2-y_1}{x_2-x_1}$$

$$\frac{y+1}{x+1} = \frac{3}{4} \quad (1)$$

$$4y+4 = 3x+3$$

$$\therefore 3x-4y-5=0 \quad (1)$$

$$iii) r = \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}$$

$$= \frac{|3(0) - 4(4) - 5|}{\sqrt{9+16}} \quad (1)$$

$$= \frac{|-21|}{5}$$

$$= \frac{21}{5} u \quad (1)$$

$$iv) Area = \frac{1}{2} ab \sin C$$

$$= \frac{1}{2} \cdot 5 \times \frac{21}{5}$$

$$= \frac{21}{2}$$

$$= 10 \frac{1}{2} u^2 \quad (1)$$

$$(13) (a) 2x^2 - 3x - 4 = 0$$

$$i) x + p = -\frac{b}{a} = \frac{3}{2} \quad (1)$$

$$ii) x + p = \frac{c}{a} = -\frac{4}{2} = -2 \quad (1)$$

$$iii) (x-p)^2 = x^2 - 2xp + p^2$$

$$= (x^2 + p^2) - 2xp$$

$$= (x+p)^2 - 2xp - 2xp$$

$$= (x+p)^2 - 4xp \quad (1)$$

$$= \frac{9}{4} - 4 \cdot -2$$

$$= 10 \frac{1}{4} \quad (1)$$

$$(b) \sin \theta = \frac{1}{\sqrt{10}} + \frac{1}{\sqrt{10}}$$

$$= \frac{\sqrt{10} + \sqrt{10}}{\sqrt{10} \cdot \sqrt{10}}$$

$$= \frac{2}{10} = \frac{1}{5} \quad (1)$$

$$(4) 2 \sin 2\theta = -\sqrt{3}$$

$$\sin 2\theta = -\frac{\sqrt{3}}{2}$$

$$B.A = 60^\circ \quad (1)$$

$$\therefore 2\theta < 0 \text{ or } 3\pi, \text{ or } \pi$$

$$\therefore 2\theta = (180^\circ + 60^\circ), (360^\circ - 60^\circ)$$

$$= (540^\circ + 60^\circ), (720^\circ - 60^\circ)$$

$$= 240^\circ, 300^\circ, 600^\circ, 660^\circ \quad (1)$$

$$\therefore \theta = 120^\circ, 150^\circ, 300^\circ, 330^\circ \quad (1)$$

$$(c) |x-3| = 3x+1$$

$$x-3 = 3x+1 \quad x-3 = -(3x+1)$$

$$-4 = 2x \quad x-3 = -3x-1$$

$$x = -2 \quad 4x = 2$$

$$x = \frac{1}{2} \quad (1)$$

$$\text{For: } x = -2 \quad \sin \theta = \frac{1}{5}$$

$$\sin \theta = \frac{1}{5} \Rightarrow \theta = \sin^{-1} \frac{1}{5}$$

$$= \sin^{-1} 0.2$$

$$\sin \theta = \frac{1}{5} \Rightarrow \theta = \sin^{-1} \frac{1}{5}$$

$$= \sin^{-1} 0.2$$

$$\therefore x = \frac{1}{2} \text{ or } x = -2 \quad (1)$$

$$(d) x^2 + kx + 4 = 0$$

$$\Delta = b^2 - 4ac$$

$$= k^2 - 4 \cdot 4$$

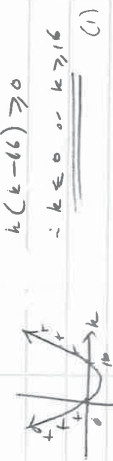
$$= k^2 - 16 \quad (1)$$

$$\text{Real roots} \rightarrow \Delta \geq 0 \quad (1)$$

$$k^2 - 16 \geq 0$$

$$k(k-4) \geq 0$$

$$\therefore k \leq 0 \text{ or } k \geq 4 \quad (1)$$



$$(e) \sin \theta = \frac{1}{\sqrt{10}} + \frac{1}{\sqrt{10}}$$

$$= \frac{\sqrt{10} + \sqrt{10}}{\sqrt{10} \cdot \sqrt{10}}$$

$$= \frac{2}{10} = \frac{1}{5} \quad (1)$$

$$= \sin^{-1} \frac{1}{5}$$

$$= \sin^{-1} 0.2$$

$$= \sin^{-1} 0.2 \quad (1)$$

$$(10) a) \quad y = (3x+1)(x-2) = 3x^2 - 5x - 2$$

$$\frac{dy}{dx} = 6x - 5 \quad (1)$$

$$\text{Slope of given line} = \frac{-7}{-1} = 7 \quad (2)$$

$$\text{We require } \frac{dy}{dx} = 7$$

$$6x - 5 = 7$$

$$6x = 12$$

$$x = 2$$

$$\text{When } x=2, y = 7 \times 0 = 0 \quad (1)$$

$$\text{Eqn of tangent: } (2, 0), m=7$$

$$y - y_1 = m(x - x_1)$$

$$y - 0 = 7(x - 2)$$

$$y = 7x - 14 \quad (1)$$

$$(11) \quad f(x) = x^2 - 3x$$

$$f(x+h) = (x+h)^2 - 3(x+h)$$

$$= x^2 + 2xh + h^2 - 3x - 3h$$

$$f(x+h) - f(x) = x^2 + 2xh + h^2 - 3x - 3h - (x^2 - 3x)$$

$$= 2xh + h^2 - 3h \quad (1)$$

$$\frac{f(x+h) - f(x)}{h} = \frac{2xh + h^2 - 3h}{h}$$

$$= 2x + h - 3 \quad (1)$$

$$f'(x) = \lim_{h \rightarrow 0} \left[ \frac{f(x+h) - f(x)}{h} \right]$$

$$= \lim_{h \rightarrow 0} [2x + h - 3]$$

$$= 2x - 3 \quad (1)$$

$$(12) \quad y = x(2x-1)^3$$

$$y' = x \cdot 3(2x-1)^2 \cdot 2 + (2x-1)^3 \quad (1)$$

$$= 6x(2x-1)^2 + (2x-1)^3$$

$$= (2x-1)^2 [6x + 2x-1] \quad (1)$$

$$= (2x-1)^2 (8x-1) \quad (1)$$

(13)

$$y = \frac{x^2}{x+1}$$

$$(i) \quad \frac{dy}{dx} = \frac{(x+1) \cdot 2x - x^2 \cdot 1}{(x+1)^2} \quad (1)$$

$$= \frac{2x^2 + 2x - x^2}{(x+1)^2}$$

$$= \frac{x^2 + 2x}{(x+1)^2} \quad (1)$$

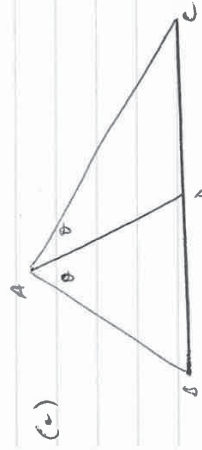
$$(ii) \quad \text{We require } \frac{dy}{dx} = 0$$

$$\therefore \frac{x^2 + 2x}{(x+1)^2} = 0$$

$$\therefore x^2 + 2x = 0$$

$$x(x+2) = 0 \quad (1)$$

$$\therefore x = -2, 0$$



$$\text{Let } \angle ABD = \alpha \quad \angle ADC = 180^\circ - \alpha$$

$$\text{In } \triangle ABD, \quad \frac{BD}{AB} = \frac{\sin B}{\sin \alpha}$$

$$\text{In } \triangle ADC, \quad \frac{DC}{AC} = \frac{\sin C}{\sin(180^\circ - \alpha)} = \frac{\sin B}{\sin \alpha}$$

$$\therefore \frac{BD}{AB} = \frac{DC}{AC}$$

$$\therefore BD \cdot AC = DC \cdot AB$$

$$\frac{BD}{DC} = \frac{AB}{AC}$$

(1) for sine rule rule

(1) for  $\sin(180^\circ - \theta) = \sin \theta$

(1) correct ratios

(14)

$$\text{In } \triangle ABD, \quad \frac{\sin B}{BD} = \frac{\sin C}{AD}$$

$$\therefore \sin B = \frac{AD \sin C}{BD} \quad \dots (1)$$

In  $\triangle ADC$

$$\frac{\sin B}{DC} = \frac{\sin C}{AD}$$

$$\sin C = \frac{AD \sin B}{DC} \quad \dots (2)$$

In  $\triangle ABC$

$$\frac{\sin B}{AC} = \frac{\sin C}{AB}$$

$$\therefore \frac{\sin C}{\sin B} = \frac{AB}{AC}$$

$$\text{From (1) and (2), } \frac{AB}{AC} = \frac{AD \sin B}{DC} \cdot \frac{AD \sin C}{BD}$$

$$= \frac{BD}{DC}$$

(1) for sine rule rule

(2) for sine rule another two times

(1) for correct ratios